

$$7.1. \quad J_{0,2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \boxed{k \in \mathbb{N}}$$

$$J_{0,2}^k = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^k \Rightarrow J_{0,2}^k = 0_{2 \times 2} \quad \text{pro } k \geq 2$$

$$J_{0,3} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$J_{0,3}^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{pro } \forall k \geq 3 \quad J_{0,3}^k = 0_{k \times k}$$

$$J_{0,5} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J_{0,5}^2 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J_{0,5}^3 = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J_{0,5}^4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\text{pro } \forall k \geq 5 : J_{0,5}^k = 0_{5 \times 5}$$

$$[g]_B^B = J - 0I_n = J = [f]_B^B$$

$$\text{pro } \forall x \in \mathbb{N} : \text{Ker } f^x \subset \text{Ker } f^{x+1}$$

$$\text{pro } \forall x \in \mathbb{N} : \text{Im } f^{x+1} \subset \text{Im } f$$

Tedy pro $\forall j < 5 \ \forall i > 3: \underbrace{\text{Ker} f^i \cap \text{Im} f^j}_{*} = \text{Im} f^j$

protože:

$$1) \text{ pro } \forall x \in \mathbb{N}: \text{Ker} f^x \subset \text{Ker} f^{x+1}$$

pro $\forall x \in \mathbb{N}, x > 3$

$$\text{jestliže } \text{Ker} f^x = \mathbb{R}^{10} \Rightarrow \text{Ker} f^{x+1} = \mathbb{R}^{10}$$

$$2) \text{ pro } \forall x \in \mathbb{N}: \text{Im} f^x \subset \mathbb{R}^{10} \Rightarrow *$$

dále pro $\forall j > 4 \ \forall i: \text{Ker} f^i \cap \text{Im} f^j = \{\vec{0}\}$

protože:

$$1) \text{ pro } \forall x \in \mathbb{N}: \text{Im} f^{x+1} \subset \text{Im} f^x$$

$$2) \text{ pro } \forall x \in \mathbb{N}: f^x(\vec{0}) = \vec{0} \Rightarrow \vec{0} \in \text{Ker} f^x \wedge \vec{0} \in \text{Im} f^x$$

